

INCIDENT INVESTIGATION REPORT

Anhydrous Ammonia Release - Refrigeration Machinery Room

Incident date: 14 March 2024 Report issued: 28 March 2024
Facility: Cold storage and food processing Location: Vijayawada, Andhra Pradesh
Classification: Fatality Report reference: INV-2024-0314

SUMMARY

On 14 March 2024 at approximately 02:40, an anhydrous ammonia release occurred in the refrigeration machinery room of a cold storage and food processing facility. A maintenance technician working alone in the machinery room was fatally injured. Two additional employees were treated for respiratory irritation and released. Approximately 180 kilograms of ammonia were released over 22 minutes before the system was isolated.

SEQUENCE OF EVENTS

At 02:31 a high-pressure alarm annunciated on the high stage compressor discharge. The control room operator acknowledged the alarm and did not dispatch anyone, as the same alarm had annunciated on eleven occasions in the preceding fourteen days and had each time cleared without intervention. At 02:38 the technician entered the machinery room to investigate a reported noise. At 02:40 the discharge line coupling failed and released ammonia into the enclosed room. The technician was not wearing respiratory protection and did not carry a gas monitor. Emergency ventilation did not start automatically. The system was isolated manually at 03:02 by a second shift supervisor who entered wearing a self-contained breathing apparatus.

IMMEDIATE CAUSES

The discharge line coupling failed under normal operating pressure. The pressure relief valve protecting the high stage discharge had been isolated by a closed block valve, leaving the line with no relief path. The technician entered an enclosed space containing a known toxic inhalation hazard without testing the atmosphere and without respiratory protection.

ROOT CAUSES

The block valve isolating the relief path was closed during a temporary repair eleven months earlier. The repair was not processed through management of change and no review of the relief system was performed. The valve was not car-sealed open and was not included in any routine inspection round, so its position was never verified again. Mechanical integrity records show wall thinning and corrosion was identified on adjacent piping during a 2022 inspection; the finding was recorded but no follow-up inspection of the discharge line was scheduled. Repeated high-pressure alarms had been normalised by operating staff and no alarm rationalisation had been carried out. The facility had no written procedure requiring atmospheric testing before entry to the machinery room.

CONTRIBUTING FACTORS

The technician was working alone on night shift with no attendant and no means of summoning assistance from inside the machinery room. Emergency ventilation was interlocked to a detector that had been out of service since a failed calibration in January 2024. Staff had not been trained on the ammonia emergency response procedure within the preceding twelve months.

EQUIPMENT INVOLVED

High stage reciprocating compressor, discharge piping and coupling, pressure relief valve and associated block valve, ammonia detection system, emergency ventilation fan.